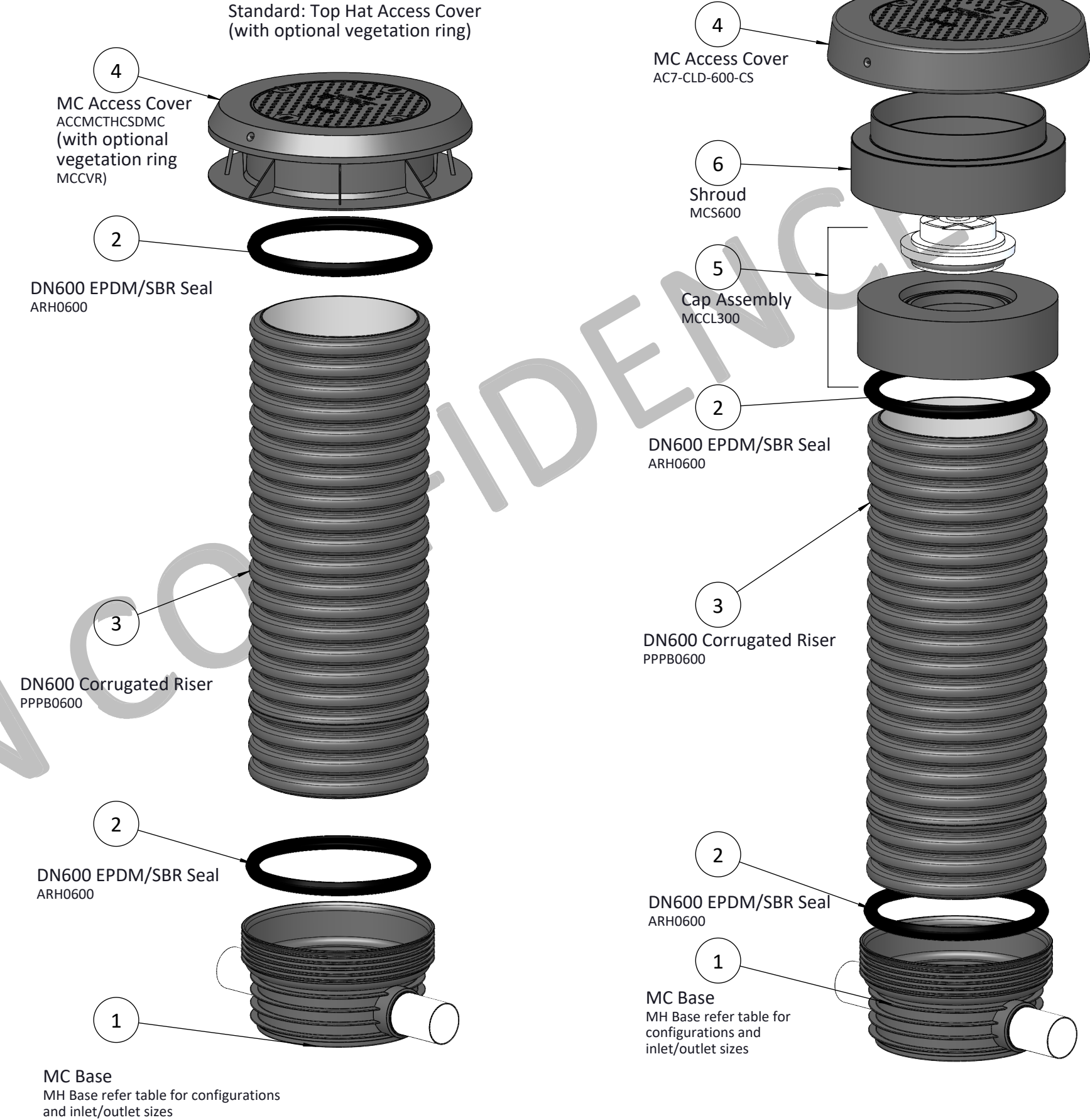
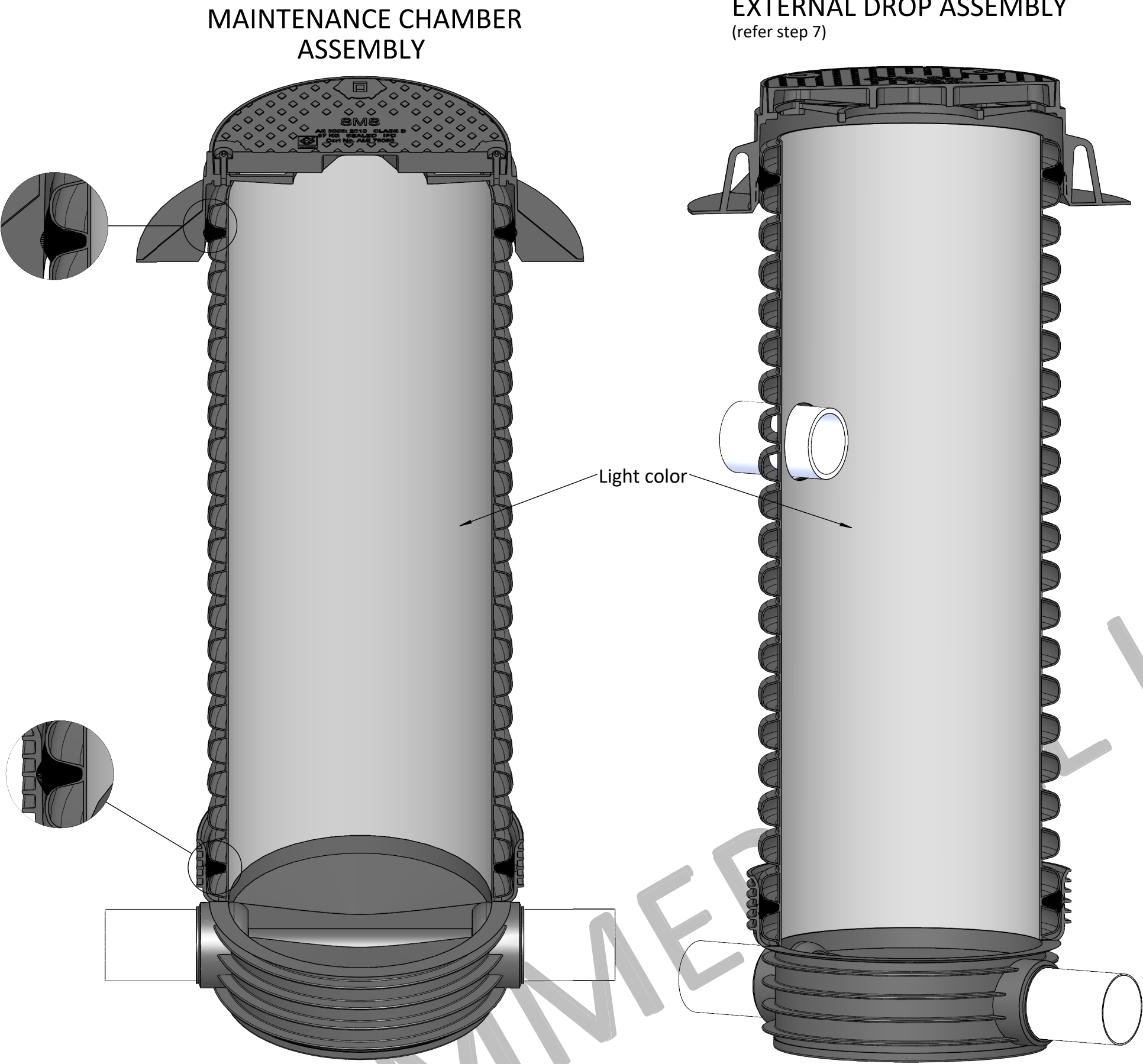


- 01 - COVER SHEET
- 02 - GA DRAWING
- 03 - ASSEMBLY AND INSTALLATION
- 04 - HEIGHT ADJUSTMENT
- 05 - INSTALLATION OVERVIEW (INCLUDING ACCESS COVER OPTIONS)
- 06 - INSTALLATION DEPTHS

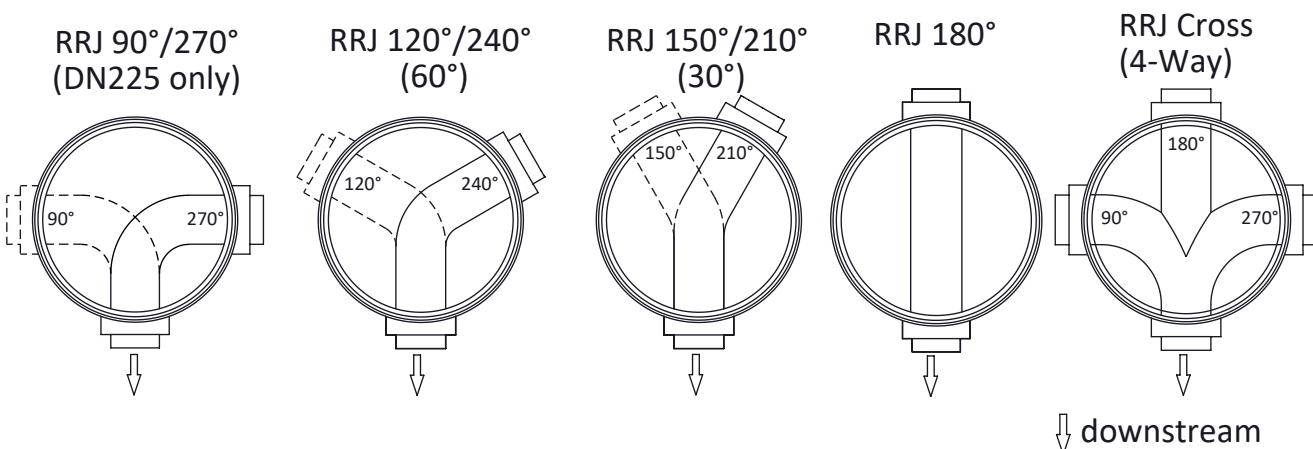
3	FLOW DIAGRAM UPDATED, ADDED IMPORTANCE NOTES		02/09/25	SB	HB
2	UPDATED SHEET 5 WITH CLEAR OPENING SIZES		09/05/25	SB	HB
1	INITIAL RELEASE		01/04/25	SB	HB
REV.	DESCRIPTION		DATE	DRAWN	APVD
REVISIONS					
E art	Material:		Description: SMS/DYKA DN600 POLYPROPYLENE (PP) MAINTENANCE CHAMBER		
	Finish:				
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			Drawing Number:		Revision:
DO NOT SCALE DRAWING IF IN DOUBT ASK			MC-ASSY-DN600-100	Sheet: 1 of 6	A
)			Drawn by: SB		Drawn Date: 01/04/25
			Checked/Approved by: HB		Checked/Approved Date: 01/04/25

SMS/DYKA DN600 POLYPROPYLENE (PP) MAINTENANCE CHAMBER (WITH 300mm CLEAR OPENING ACCESS COVER)
(INCLUDING INSTALLATION GUIDELINES)



AVAILABLE STANDARD CONFIGURATIONS

MC Base		
CODE	CONFIGURATION	INLET/OUT SIZE
AXEDO 150/120	RRJ 60° / 120° / 240°	DN150
AXEDO 150/150	RRJ 30° / 150° / 210°	DN150
AXEDO 150/180	RRJ 180° Straight	DN150
AXEDO 150/SX	RRJ Cross (4-Way)	DN150
AXEDO 225/90	RRJ 90° / 270°	DN225
AXEDO 225/120	RRJ 60° / 120° / 240°	DN225
AXEDO 225/150	RRJ 30° / 150° / 210°	DN225
AXEDO 225/180	RRJ 180° Straight	DN225
AXEDO 225/SX	RRJ Cross (4-Way)	DN225



INDUSTRY STANDARDS

WSA 137 – Industry Standard for Unplasticised Poly (vinyl chloride) (PVC-U), Polypropylene (PP), and Polyethylene (PE) Maintenance Shafts, Chambers, and Holes for Sewerage
EN 13598-2 – Plastics Piping Systems for Non-Pressure Underground Drainage and Sewerage – Specifications for Manholes and Inspection Chambers in Traffic Areas and Deep Underground

Installations

AS/NZS 2566 for Buried Flexible Pipelines
AS/NZS 5065 – Polyethylene and Polypropylene Pipes and Fittings for Drainage and Sewerage Applications
AS 3996 – Access Covers and Grates
AS 1646 – Elastomeric Seals for Waterworks Purposes

APPLICABLE PRODUCT SPECIFICATIONS

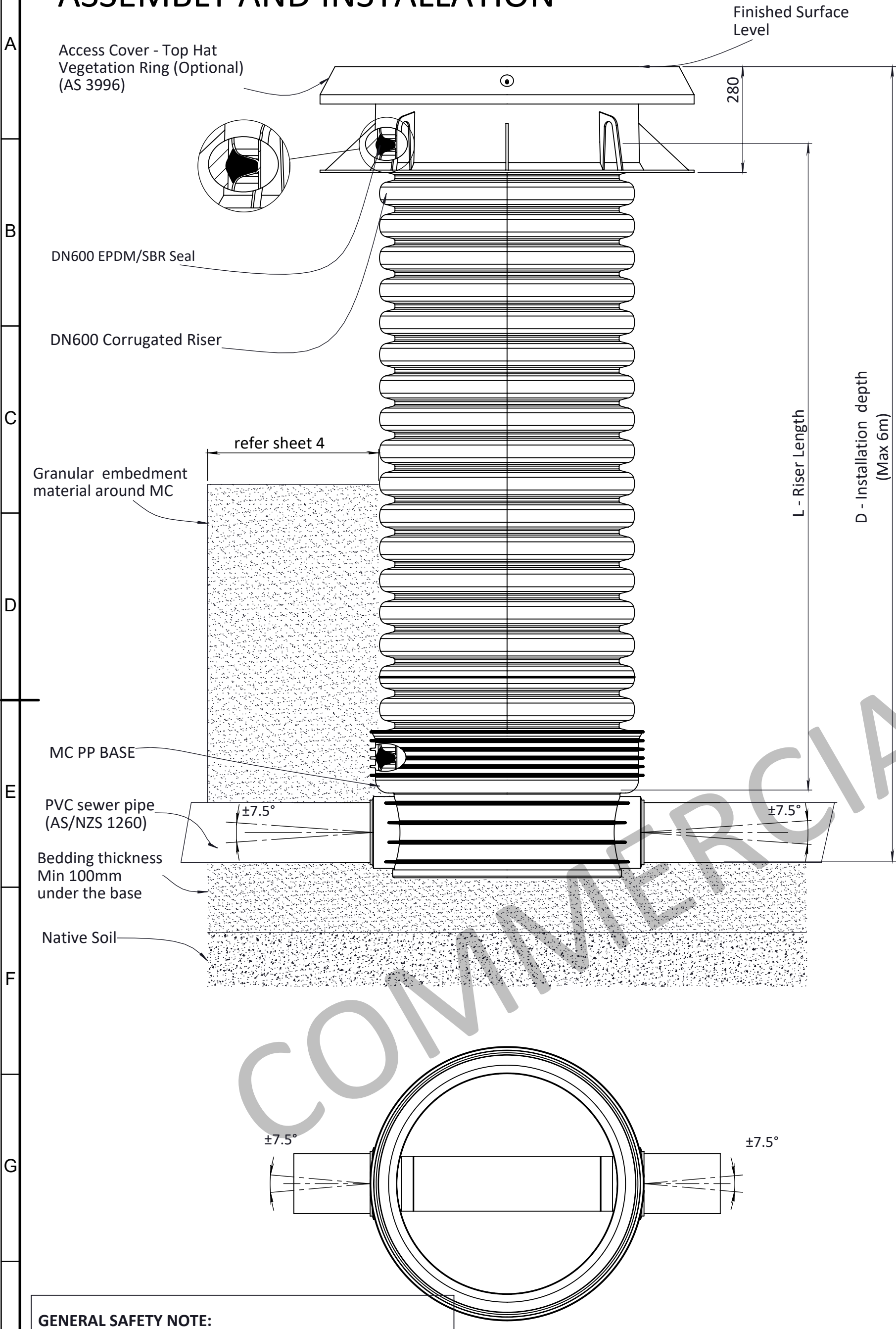
WSA PS-337: Maintenance Chambers (MC) – Polypropylene (PP) for Non-Pressure Applications – Gravity Sewerage
WSA PS-290 – Access Covers and Frames for Water Supply and Sewerage (WSA 132 Compliance)
WSA PS-230 – Unplasticised Polyvinyl Chloride (PVC-U) Pipes and Fittings for Non-Pressure Sewerage and Drainage Applications

SITE SAFETY

Carry out relevant safety assessment/s before carrying out any installation/construction activities.

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		Finish:	Part Number: -	
Approx Weight:		Drawing Number: MC-ASSY-DN600-100		Sheet: 2 of 6
Drawing Scale: NTS		DO NOT SCALE DRAWING IF IN DOUBT ASK		Revision: A
Drawing Produced In Accordance With: AS 1100		Drawn by: SB		Drawn Date: 01/04/25
Projection Method: THIRD ANGLE		Sheet Size: A2		Checked/Approved by: HB
				Checked/Approved Date: 01/04/25

ASSEMBLY AND INSTALLATION



GENERAL SAFETY NOTE:
All installation, handling, and maintenance activities must be carried out in strict accordance with relevant local, national, and industry safety standards and regulations. Appropriate personal protective equipment (PPE) must be worn at all times.

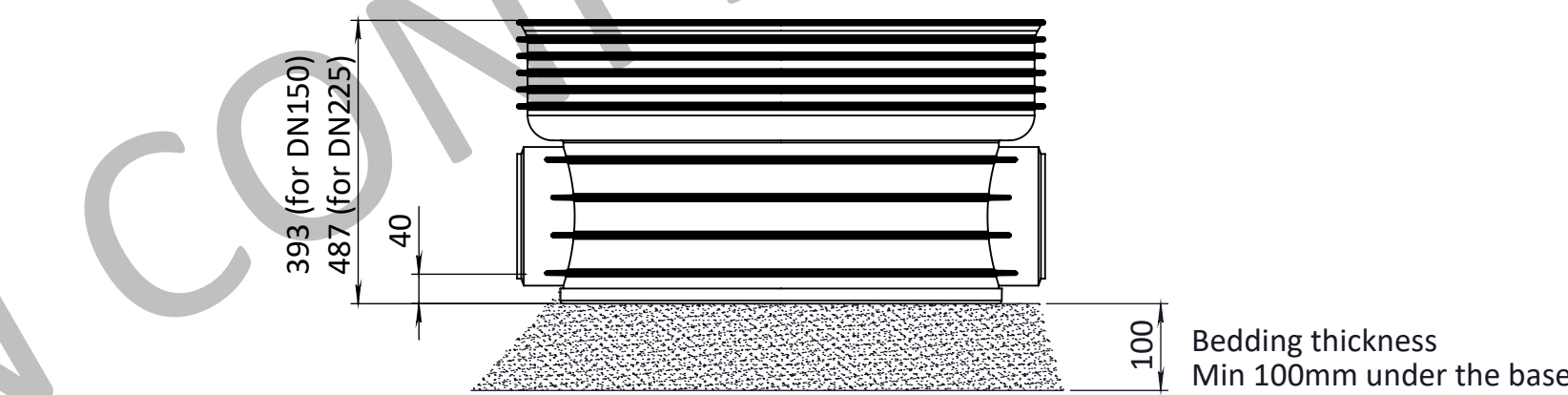
- 1. TRANSPORT AND STORAGE**
Store maintenance chamber components on level ground. If stored outdoors for an extended period, protect the chambers from direct sunlight. All supplied seals must remain in their original packaging and be stored in a dry, protected environment.
- 2. GENERAL INFORMATION**
SMS/DYKA PP maintenance chambers are supplied ready for connection. Upon delivery, check all components for completeness and inspect them for any damage or contamination. Damaged components must not be installed.

- 3. ASSEMBLY AND INSTALLATION**
All of the installation parameters listed below must be followed and as detailed in this installation guide.

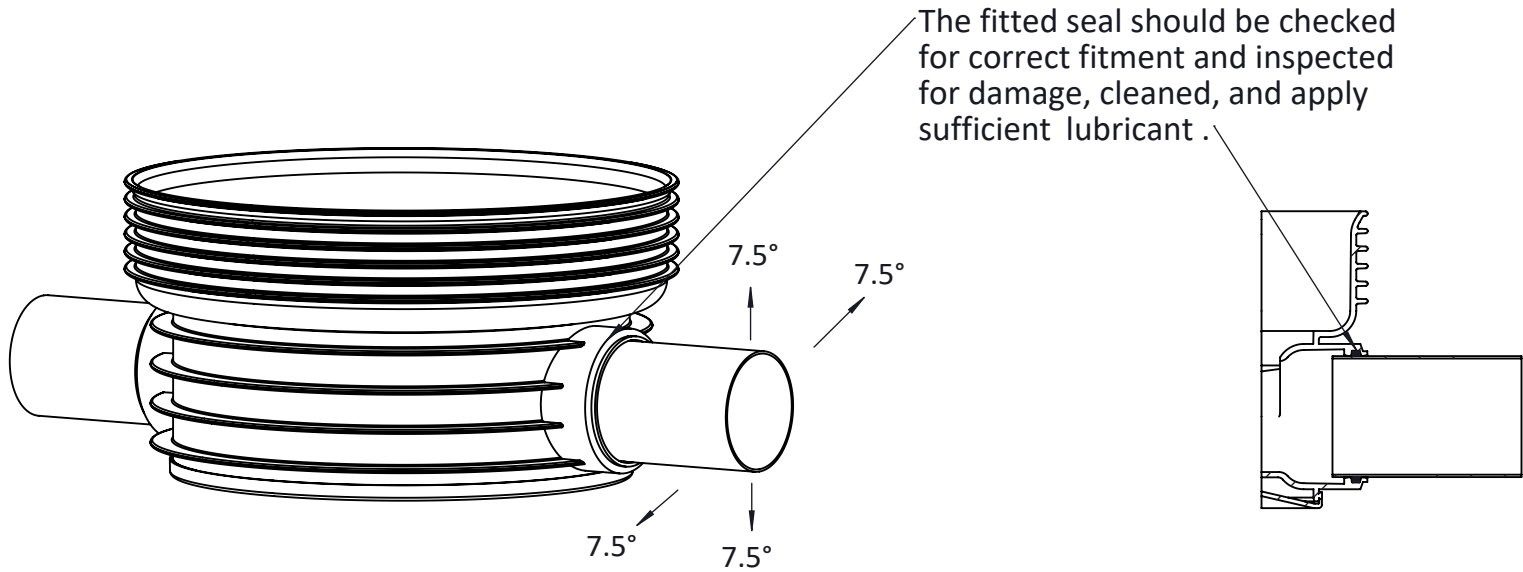
Step 1: BEDDING (GRANULAR SUB-BASE)
The minimum depth required below the base is 100 mm. The support area of the maintenance chamber base must be load-bearing and levelled.

The support area of the maintenance chamber base must be in accordance with the approved installation plans.

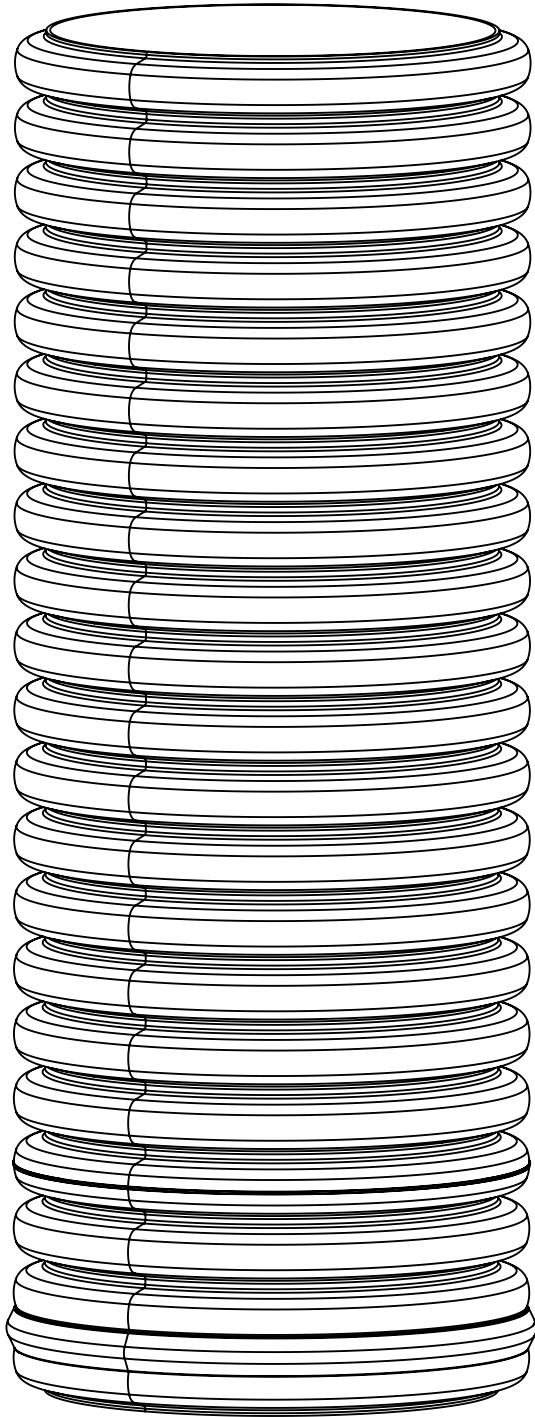
Step 2: BASE/PIPE CONNECTION
The chamber base shall be positioned and levelled on the prepared support area, in alignment with the connecting pipes. The adjustment and flow direction of the maintenance chamber base must be checked.



Step 2.1 PP-BASE WITH SOCKETS
The socket seals should be checked for correct fitment and inspected for any damage; cleaning may be necessary. Apply sufficient lubricant to the the socket pipe seal as well as at the end of the spigot and fully inserting the spigot end into the socket. Ensure pipe spigot is correctly chamfered. (Suggested approved lubricant: Thomas Grozier BK Bactericidal Pipe Lubricant.)
For all base sockets, horizontal angles of $\pm 7.5^\circ$ and gradient changes up to $\pm 7.5^\circ$ are achievable. Simultaneous direction and gradient changes will reduce the indicated maximum values accordingly. No connectors (short pipes or joints) are required between SMS/DYKA PP maintenance chamber and pipes, unless specified by local water agency.



Step. 3 CONNECTION OF MAINTENANCE CHAMBER COMPONENTS
Ensure the pipe and first pipe corrugated valley are clean. Fit the seal in first corrugated valley and apply lubricant. Then, clean and lubricate the interior surface of the base before fitting the riser securely into position.



Seal (ARH0600) is slipped on to pipe corrugated valley.
Once fitted, thoroughly clean the seal and apply sufficient lubricant.

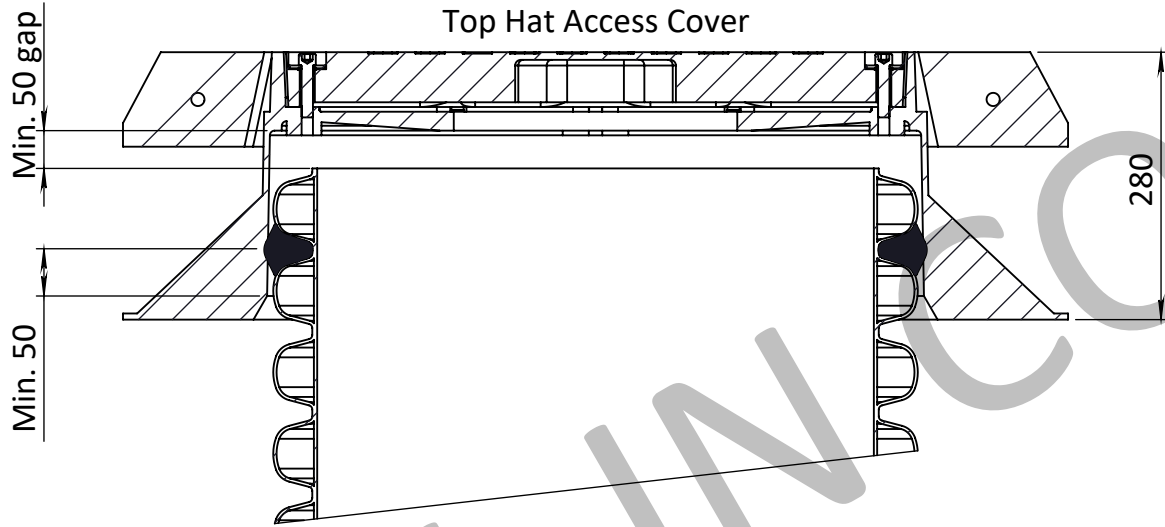
Step 4: BACKFILLING MATERIAL
The bedding shall be granular material, embedment sand (WSA-PS-360) or 5mm minus crushed rock (WSA-PS-361).

Step 5: BACKFILLING AND COMPACTING
The trench width for backfilling around the maintenance chamber must be in at any point at least 400 mm. When installing the maintenance chambers in high water table/groundwater conditions (refer local water agency specifications), the trench width of at least 500 mm is to be maintained all around to prevent uplift.

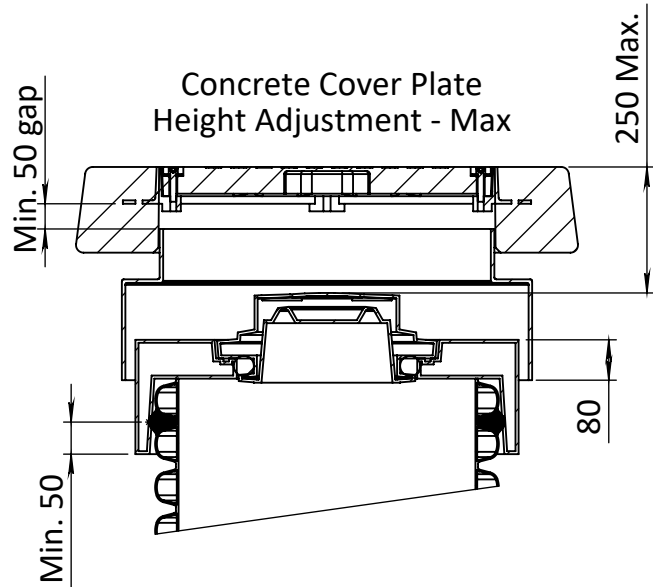
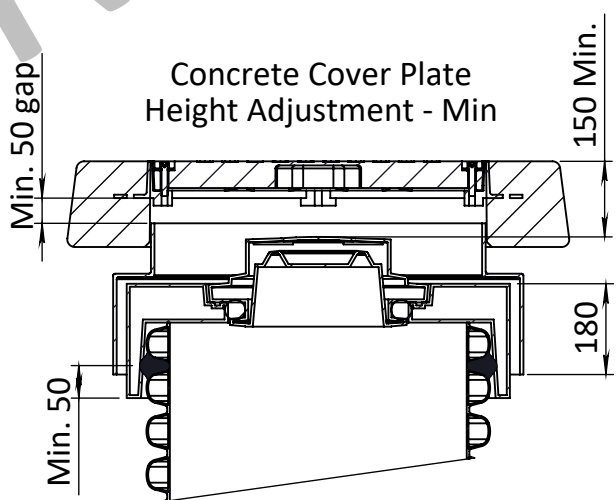
The area around the pipe connection to the maintenance chamber must be carefully under-packed/haunched to ensure correct pipe support e.g., using a narrow hand stamper. The backfilling material should be inserted carefully and in layers of 200-400 mm layer thickness and compacted with a medium vibrating stamper (approx. 50kg).

The number of required compacting passes per layer depends on the backfilling material. A minimum degree of compaction of DPr = 97 % is to be established for the entire depths of the maintenance chamber.

Step 6.1: TOP HAT ACCESS COVER HEIGHT ADJUSTMENT
Trim the corrugated pipe to the required height, ensuring a minimum 50mm gap between the top hat access cover and the corrugated pipe (refer below and Sheet 5 for corrugated pipe height calculations). The cut must be made on the corrugation valleys. Finish by deburring the cut surface.



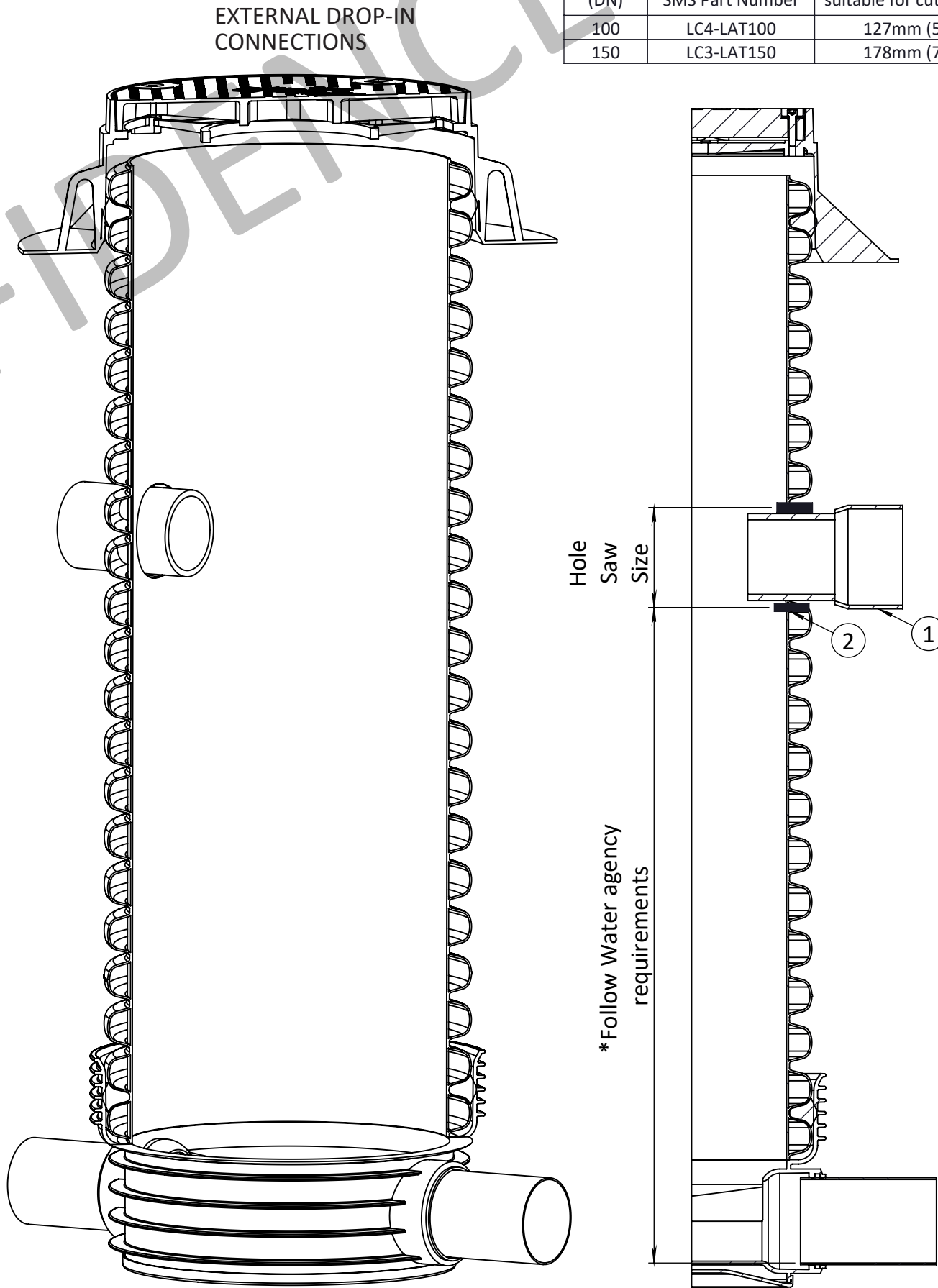
Step 6.2: CONCRETE COVER PLATE HEIGHT ADJUSTMENT
Trim the corrugated pipe to the required height, ensuring a minimum 50mm gap between the access cover and the shroud (refer below and Sheet 5 for corrugated pipe height calculations). The cut must be made on the corrugation valleys. Finish by deburring the cut surface.



Step 7: DN100/DN150 LATERAL DROP-IN CONNECTIONS (OPTIONAL -INSTALL ON-SITE)
Using an electric hand drill with a hole saw cutter, drill from the inside at the desired location, ensuring that the drilling depth does not exceed the specified maximum limit. Do not drill in any area where a connecting seal is present. After drilling, carefully deburr the hole to remove sharp edges and debris. Insert the seal from the outside of the maintenance chamber, making sure the collar of the seal sits flush against the ribs on the outer surface.

Lubricate both the spigot end of the pipe and the inner surface of the seal before inserting the pipe. Carefully push the pipe into the seal, ensuring it creates a secure inner overlap for a proper connection. For multiple drop-in connections, repeat same procedure as permitted by local water agency, adhering to their specific installation guidelines for such connections.

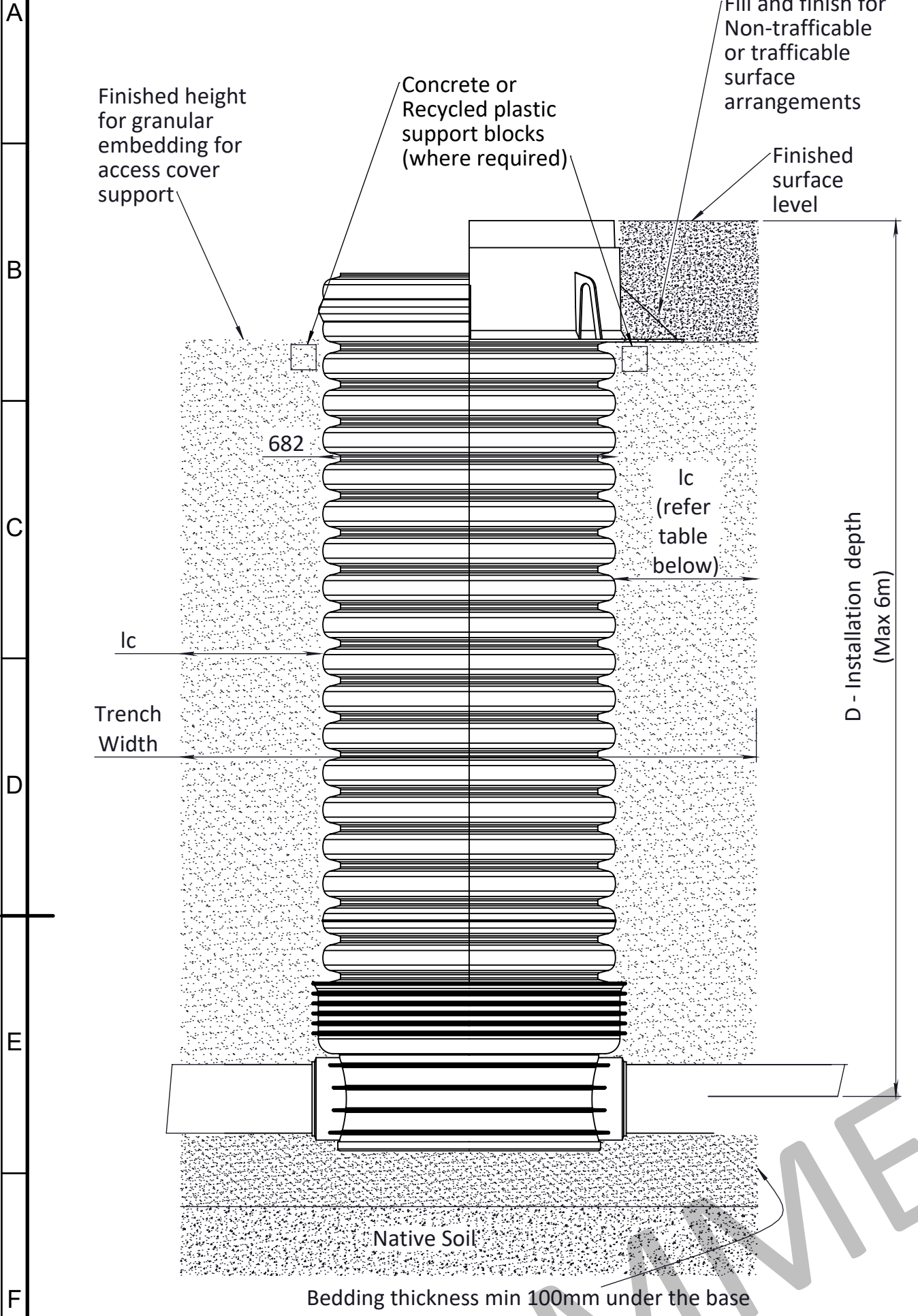
Inlet Pipe (DN)	Lateral Connector SMS Part Number	Hole Saw Cutter Size suitable for cutting PP
100	LC4-LAT100	127mm (5")
150	LC3-LAT150	178mm (7")



ITEM NO.	DESCRIPTION	MATERIAL
1	DN100/DN150 Straight Connector	AS/NZS 1260 PVC
2	DN100/DN150 Rubber Collar	AS681.1 Rubber

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		Finish:		Part Number:	
Approx Weight:		Drawing Number:		Sheet:	Revision:
Drawing Scale: NTS		DO NOT SCALE DRAWING IF IN DOUBT ASK		MC-ASSY-DN600-100	4 of 6 A
Drawing Produced In Accordance With: AS 1100		Drawn by:		Drawn Date:	
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HB		01/04/25			

INSTALLATION OVERVIEW (INCLUDING ACCESS COVER OPTIONS)



- Installation notes:**
- Refer to local Water Agency specifications for other specific installation guidelines.
 - For installation in reactive soils, refer to local water agency installation guidelines.
 - Concrete used for installation or concrete components to comply to local water agency specifications.
 - Test installed maintenance chamber to local water agency requirements.
 - Refer to AS/NZS 2566 for Buried Flexible Pipelines and local authority requirements for any specific installation requirements.
- Bedding, Backfilling/Embedment material:**
- The bedding shall be granular material, embedment sand (WSA-PS-360) or 5mm minus crushed rock (WSA-PS-361).
 - Compaction: DPr ≥ 97%.

Trench widths	Ic (Min.)	Trench Width (Min.)
Standard Installation	400	1482
Groundwater conditions	500	1682

- Installation depth notes:**
- Standard Installation max 6m depth, in groundwater conditions max 5m depth installation.

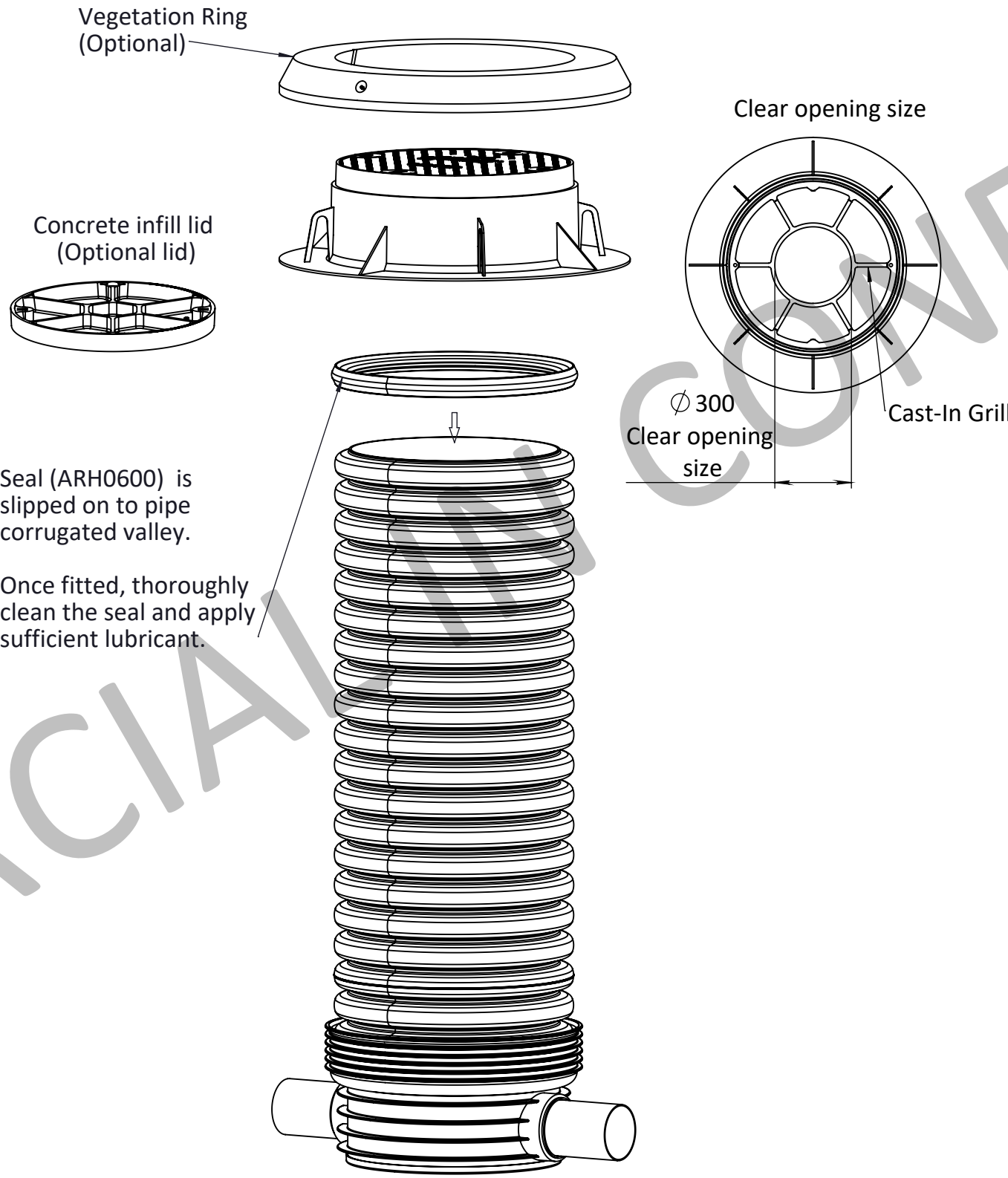
STANDARD: TOP HAT ACCESS COVER (OPTIONAL VEGETATION RING)

The SMS Class D Top hat access cover is designed to distribute loads to the surrounding foundation, preventing direct load transfer to the PP maintenance chamber.

Before installation, fit the seal ARH0600 in the first corrugated valley of the pipe riser (the surface is thoroughly cleaned and apply sufficient lubricant). Assemble the Top hat access cover horizontally and it is centered over the maintenance chamber on a stable, prepared base.

Important: Ensure there is no direct load contact between the access cover and the PP maintenance chamber.

The bedding for the Top hat access cover must be level and free from point loads, using suitable materials such as granular fill, sand, or concrete as needed.



Upon completion of the installation and any required testing, the access cover lid and bolting need to be removed safely. Apply DENSO Manhole sealing grease to the access cover seating area. Reinstall the access cover lid and tighten the bolts to 80-100 Nm. Finally, re-fit the plastic caps over the bolts.

IMPORTANT NOTE: Follow all installation guides including minimum gap allowances to ensure any loads are not directly transferred to the riser pipe.

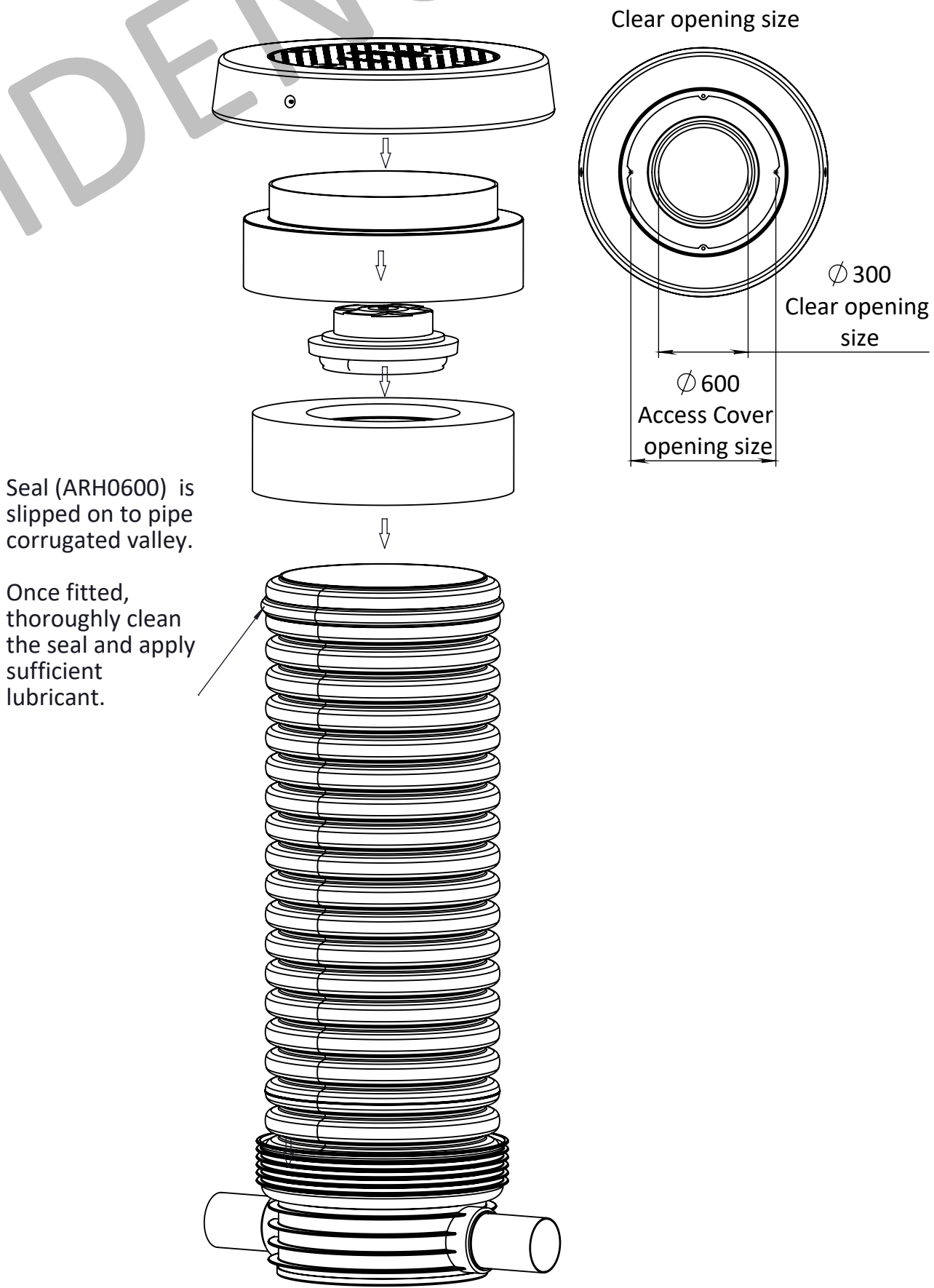
OPTION: CONCRETE COVER PLATE

The SMS Class D concrete cover plate is designed to distribute loads to the surrounding foundation, preventing direct load transfer to the PP maintenance chamber.

Before installation, fit the seal ARH0600 in the first corrugated valley of the pipe riser (the surface is thoroughly cleaned and apply sufficient lubricant). Push the cap assembly fully into place, then position the shroud over the riser pipe and cap assembly. Assemble cover plate horizontally and centered over the maintenance chamber on a stable, prepared base. Below the cover plate, place support blocks on either side of the shroud as shown in Sheet 5.

Important: Ensure there is no direct load contact between the cover plate and the PP maintenance chamber.

The bedding for the cover plate must be level and free from point loads, using suitable materials such as granular fill, sand, or concrete as needed. The cover plate must be centered during installation without disturbing the bedding.



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